Calc 1: Special Technique & Review

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DIRECTIONS

Problems 1-6 use logarithmic differentiation (Special Technique): take ln of both sides, differentiate implicitly, then solve for dy/dx. Problems 7-10 cover normal lines, limit definition of derivative, simplification, and domain/range.

1 Find dy/dx if:

$$y = 3^{-x}$$

Answer: ________________________

2 Find dy/ds if:

$$y = 2^{s^2}$$

Answer: ________________________

3 Find dy/dt if:

$$y = 2^{\sin 3t}$$

Answer: ________________________

4 Find dy/dx if:

$$y = (x+1)^x$$

Answer: ________________________

5 Find dy/dx if:

$$y = x^{x+1}$$

Answer: ________________________

6 Find dy/dx if:

$$y = x^{\sin x}$$

Answer: ________________________

7 Find the normal line at (1, 1):

$$f(x) = x^3$$

Answer: ________________________

8 Evaluate the limit:

$$\lim_{h \to 0} \frac{5(x+h)^3 - 5x^3}{h}$$

Answer: ________________________

9 Simplify:

$$e^{\ln 2 + \ln x^2}$$

Answer: ________________________

10 Find domain and range of:

$$y = -\sqrt{-x}$$

Answer: ________________________

Answer Key & Solutions

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TEACHER NOTES

Problems 1-6: ln both sides, differentiate, multiply by original function. #7: f'(1)=3 (slope of tangent); normal slope = -1/3. #8: limit definition gives d/dx[5x^3] = 15x^2. #9: e^(ln2 + lnx^2) = 2x^2. #10: need -x >= 0, so x <= 0.

1 Find dy/dx if:

$y = 3^{-x}$

= $\frac{dy}{dx} = -\ln(3)\cdot 3^{-x}$

2 Find dy/ds if:

$y = 2^{s^2}$

= $\frac{dy}{ds} = \ln 2 \cdot 2s \cdot 2^{s^2}$

3 Find dy/dt if:

$y = 2^{\sin 3t}$

= $\frac{dy}{dt} = 3\ln 2 \cdot \cos(3t) \cdot 2^{\sin 3t}$

4 Find dy/dx if:

$y = (x+1)^x$

= $\frac{dy}{dx} = \left[\ln(x+1) + \frac{x}{x+1}\right](x+1)^x$

5 Find dy/dx if:

$y = x^{x+1}$

= $\frac{dy}{dx} = \left[\frac{x+1}{x} + \ln x\right]x^{x+1}$

6 Find dy/dx if:

$y = x^{\sin x}$

= $\frac{dy}{dx} = \left(\frac{\sin x}{x} + \ln x \cos x\right)x^{\sin x}$

7 Find the normal line at (1, 1):

$f(x) = x^3$

= $y = -\frac{1}{3}x + \frac{4}{3}$

8 Evaluate the limit:

$\lim_{h\to 0}\frac{5(x+h)^3 - 5x^3}{h}$

= $15x^2$

9 Simplify:

$e^{\ln 2 + \ln x^2}$

= $2x^2$

10 Find domain and range of:

$y = -\sqrt{-x}$

= $D:(-\infty, 0],\quad R:(-\infty, 0]$